

PEAK ACHIEVERS

MATHEMATICS

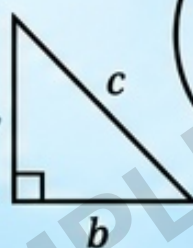
Grade 8

Book 1

Learner's Book for Competence-Based Learning

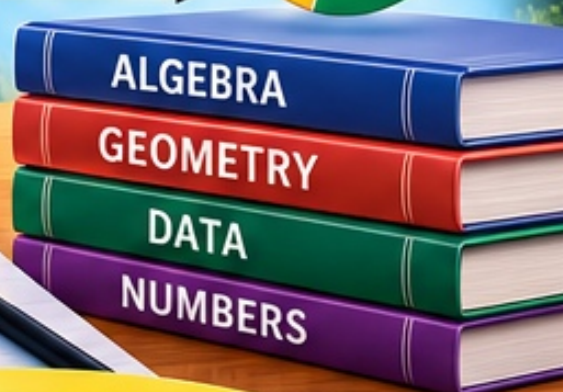
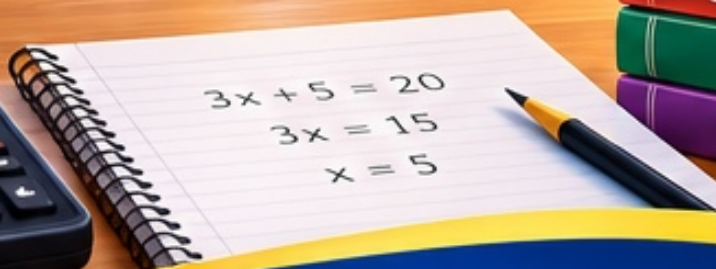
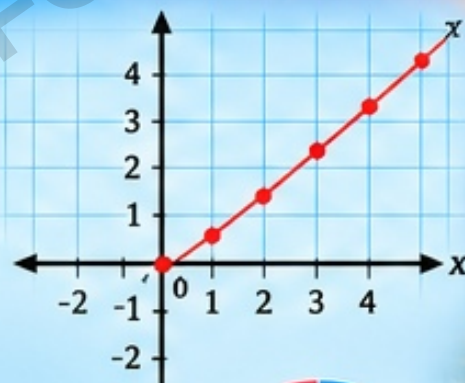


$\frac{3}{4}$



$$C = 2\pi r$$

$$A = \pi r^2$$



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PEAK ACHIEVERS ONLINE ACADEMY
MATHEMATICS - GRADE 8

BOOK 1

CONCEPTS, APPLICATION & REVISION

Aligned to Kenya's Competency-Based Education approach

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LEARN • APPLY • THINK BIG • ACHIEVE

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Dedication

In loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifice and inspiration remain part of my educational journey.

With heartfelt appreciation to Hyrine Kerubo, Peter Munde Mong'are, Teresa Kwamboka Sawe, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and all those whose support and encouragement have strengthened this journey.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart. ❤️

To my fellow teachers, guardians, family and every learner who continues to inspire the work behind Peak Achievers.

May every learner using this book remain curious, work with determination, and discover the confidence to learn, apply, think big and achieve.

LEARN • APPLY • THINK BIG • ACHIEVE

Preface

This book supports Grade 8 learners in building mathematical understanding through brief explanations, worked examples, practical situations and competency-based application. The emphasis is on understanding why a method works, choosing suitable mathematics for familiar situations, communicating working clearly and checking whether answers are reasonable.

The learning journey is: LEARN → PRACTISE → APPLY → THINK BIG → ASSESS → ACHIEVE.

How to Use This Book

- Learn the key idea before attempting questions.
- Follow worked examples carefully, then reproduce the steps independently.
- Use REMEMBER, HINT, SPOTLIGHT, ERROR DETECTIVE, EXAM SMART and THINK BIG callouts to strengthen understanding.
- Complete practical activities and real-life applications.
- Attempt each Application Quiz and Strand Revision before checking answers.
- Finish with the Mixed Exercises and five Final Model Tests.
- Use the answer and marking guidance to identify areas for improvement.

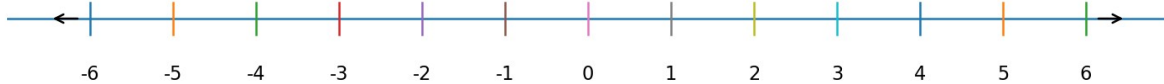
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- STRAND 2: ALGEBRA - 2.1 Algebraic Expressions; 2.2 Linear Equations
- STRAND 3: MEASUREMENTS - 3.1 Circles; 3.2 Area; 3.3 Money
- STRAND 4: GEOMETRY - 4.1 Geometrical Constructions; 4.2 Coordinates and Graphs; 4.3 Scale Drawing; 4.4 Common Solids
- STRAND 5: DATA HANDLING AND PROBABILITY - 5.1 Data Presentation and Interpretation; 5.2 Probability
- REVISION & ASSESSMENT - Strand Revisions; Mixed Exercises 1-5; Final Model Tests 1-5; Answers and Marking Guidance

STRAND 1 - NUMBERS

1.1 Integers

Figure: Integer number line



Integers are whole numbers, their negatives and zero. They can be represented on a number line.

On a number line, numbers increase to the right and decrease to the left. A negative number farther from zero is smaller.

Addition and subtraction of integers can be modelled using movement on a number line. For multiplication and division, like signs give a positive answer while unlike signs give a negative answer.

Worked Example: Evaluate $-7 + 12 - 5$. $-7 + 12 = 5$, then $5 - 5 = 0$.

REMEMBER: The sign belongs to the number. Take care when subtracting a negative number.

SPOTLIGHT: Integers are used in temperatures, elevations, financial changes and scores.

ERROR DETECTIVE: Check whether you have changed subtraction of a negative number into addition.

Application Quiz

1. Evaluate $-15 + 8$.
2. Arrange $-6, 4, -2, 0, 7$ in ascending order.
3. A temperature rises from -4°C to 7°C . Find the rise.
4. Evaluate -6×9 .
5. Explain why $(-8) \div (-2)$ is positive.

Application Quiz Answers

1. -7

2. $-6, -2, 0, 4, 7$

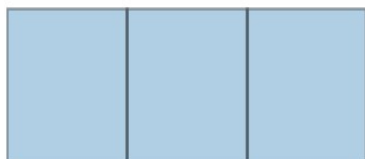
3. 11°C

4. -54

5. A negative divided by a negative gives a positive result.

1.2 Fractions

Figure: Fraction model



$\frac{3}{8}$ shaded

Fractions represent parts of a whole or quantities. Equivalent fractions have the same value.

To add or subtract unlike fractions, first use a common denominator. To multiply fractions, multiply numerators and denominators then simplify. To divide by a fraction, multiply by its reciprocal.

A fraction of a quantity is found by multiplying the fraction by the quantity.

Worked Example: $\frac{3}{4} + \frac{5}{8} = \frac{6}{8} + \frac{5}{8} = \frac{11}{8} = 1 \frac{3}{8}$.

HINT: Simplify answers to their lowest terms.

APPLICATION: If $\frac{3}{5}$ of 250 seedlings survive, the number surviving is $\frac{3}{5} \times 250 = 150$.

Application Quiz

6. Simplify $\frac{18}{24}$.
7. Calculate $\frac{7}{8} - \frac{3}{16}$.
8. Find $\frac{2}{3}$ of 360.
9. Calculate $\frac{3}{5} \div \frac{9}{10}$.
10. A tank is $\frac{3}{4}$ full. If its full amount is represented by 800 equal units, how many units are represented?

Application Quiz Answers

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MATHEMATICS - GRADE 8 - BOOK 1

FREE SAMPLE PACK

This free preview gives learners, parents and teachers a taste of the Peak Achievers revision approach: clear explanations, worked examples, application practice and competency-based assessment.

WHAT YOU GET IN THE FULL BOOK

- Book 1: Concepts, Application & Revision
- Clear Grade 8 Mathematics notes and worked examples
- Application quizzes and competency-based revision
- Full Book 1 price: KSh 350

SPECIAL BUNDLE: Book 1 + Book 2 = KSh 500

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